Part 4: Reflection & Workflow Diagram

Reflection

Question 1: What was the most challenging part of the workflow? Why?

Most Challenging: Data Preprocessing and Class Imbalance

Why:

- **Missing Data**: Healthcare data has many missing values. Choosing between imputation methods is difficult - wrong choice introduces bias
- **Class Imbalance**: Only 15-20% readmissions. Models can achieve 85% accuracy by always predicting "no readmission" but miss all high-risk patients
- **No Clinical Expertise**: As a data scientist, I don't know which features are clinically meaningful. Need constant collaboration with doctors
- **Ethical Risk**: Every preprocessing decision can introduce bias affecting patient outcomes

Example: Spent 2 hours debugging duplicate column names that caused model failures. Data quality issues can derail the entire workflow.

Why It Matters: "Garbage in, garbage out" - Poor data preprocessing creates biased models that no amount of tuning can fix.

Question 2: How would you improve your approach with more time/resources?

Key Improvements:

1. Data Quality (Highest Priority)

- Comprehensive data audit with clinicians
- Standardized data collection protocols
- Integrate external data (social determinants, pharmacy records)

2. Clinical Collaboration

- Regular meetings with physicians and nurses

- Validate all features with medical experts
- Clinical literature review for known risk factors

3. Fairness Analysis

- Evaluate performance across all demographic groups
- Quarterly bias audits
- Conduct randomized controlled trials

4. Advanced Models

- Extensive hyperparameter tuning
- Test ensemble methods
- Access to GPU for deep learning experiments

5. Robust Deployment

- Build user-friendly clinical dashboards
- 24/7 monitoring and support
- Automated retraining pipelines

Priority: Focus on data quality and clinical validation first, advanced models last.

Diagram

AI Development Workflow

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┌────────────────────────────────────────────────────────┐
│         AI DEVELOPMENT WORKFLOW                        │
│    Hospital Readmission Prediction System              │
└────────────────────────────────────────────────────────┘


┌────────────────────────────────────────────────────────┐
│ 1. PROBLEM DEFINITION                                  │
│ • Define problem & objectives                          │
│ • Identify stakeholders                                │
│ • Set KPIs (≥80% accuracy)                             │
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┌────────────────────────────────────────────────────────┐
│ 2. DATA COLLECTION                                     │
│ • Extract from EHR, billing, SDOH                      │
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• Integrate multiple sources

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3. EXPLORATORY DATA ANALYSIS
• Examine data structure
• Visualize distributions
• Identify missing values & outliers

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4. DATA PREPROCESSING
• Handle missing values
• Encode categorical variables
• Address class imbalance

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5. FEATURE ENGINEERING
• Create derived features
• Scale numerical features
• Select important features

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6. DATA SPLITTING
• Training (80%) / Test (20%)
• Stratified split

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7. MODEL TRAINING
• Train multiple models
• Random Forest selected (best performance)

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8. HYPERPARAMETER TUNING
• Cross-validation
• Optimize for F1 score

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┌──────────────────────────────────────────────────────┐
│ 9. MODEL EVALUATION                                  │
│ • Calculate metrics                                  │
│ • Create confusion matrix                            │
│ • Check fairness across demographics                 │
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              ┌───────────────┐
              │ Performance   │
              │ ≥80% accuracy?│
              └───────────────┘
                 │         │
                YES        NO
                 │         │
                 │         └──→ Return to Stage 5 or 7
                 ↓
┌──────────────────────────────────────────────────────┐
│ 10. MODEL INTERPRETATION                             │
│ • Generate SHAP explanations                         │
│ • Validate with clinicians                           │
│ • Assess fairness                                    │
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│ 11. DEPLOYMENT                                       │
│ • Create API                                         │
│ • Integrate with EHR                                 │
│ • HIPAA compliance                                   │
│ • Train staff                                        │
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│ 12. MONITORING & MAINTENANCE                         │
│ • Track performance monthly                          │
│ • Monitor for drift                                  │
│ • Retrain quarterly                                  │
└──────────────────────────────────────────────────────┘
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              ┌───────────────┐
              │ Performance   │
```

```
         │ degradation? │
         └──────────────┘
            │         │
           YES        NO
            │         │
            │         └──→ Continue Monitoring
            │
            └──→ Return to Stage 3 or 7
```

Key Principles

- **Iterative**: Loop back to earlier stages as needed
- **Collaborative**: Engage clinicians throughout
- **Ethical**: Fairness checks at every stage
- **Monitored**: Continuous oversight post-deployment

Summary

The workflow requires technical skills, clinical collaboration, ethical vigilance, and continuous improvement to create AI systems that genuinely improve patient outcomes.